

ENR 1.7 ALTIMETER SETTING PROCEDURES

1. Introduction

The Altimeter Setting Procedures in use in the Philippines generally conform to those contained in ICAO Doc 8168-OPS/311. Pertinent procedures are reproduced below. Transition altitudes are given on the corresponding approach charts in the AD section of this AIP.

2. Basic altimeter setting procedures

2.1 System of Flight Levels (FL)

2.1.1 Flight levels shall be numbered according to the following table, which indicates the corresponding height in the Standard Atmosphere, in feet:

Flight Level Number	Height in Standard Atmosphere		Flight Level Number	Height in Standard Atmosphere	
	Feet	Meters		Feet	Meters
10	1000	300	50	5000	1500
15	1500	450	---	---	---
20	2000	600	100	10000	3050
25	2500	750	---	---	---
30	3000	900	150	15000	4550
35	3500	1050	---	---	---
40	4000	1200	200	20000	6100
45	4500	1350	---	---	---
			500	50000	15250

NOTE: The height shown in meters are approximate equivalent taken from the table of cruising levels in Appendix 3 of Annex 2.

2.2 Transition Altitudes (TA)

2.2.1 A transition altitude is normally specified for each aerodrome where the aerodrome is located.

2.2.2 As far as possible, a common transition altitude is established for two or more closely spaced aerodromes. This common transition altitude is the higher of the transition altitudes that would result for the aerodrome if separately considered.

2.2.3 The height above the aerodrome of the transition is not less than 1500FT (450M).

2.2.4 The calculated height of the transition altitude is rounded up to the next full 1000FT (300M).

2.3 Transition Levels (TL)

2.3.1 When a transition level is not specified for a particular aerodrome or area, the following procedures shall be used in determining the transition level when the QNH is available:

- a. When QNH value is between 28.92 and 29.41 inches, the TL is equal to TA plus 2000FT and expressed in FL.

Example:

QNH = 29.10 inches

TA = 5000FT

TL = 5000FT + 2000FT = FL70

- b. When QNH value is between 29.42 and 29.91 inches, the TL is equal to TA plus 1500FT and expressed in FL.

Example:
QNH = 29.90 inches
TA = 5000FT
TL = 5000FT + 1500FT = FL65

- c. When QNH value is equal or greater than 29.92 inches (standard), the TL is equal to TA plus 1000FT and expressed in FL.

Example:
QNH = 29.92 inches
TA = 5000FT
TL = 5000FT + 1000FT = FL60

- 2.3.2 As far as possible, a common transition level is established for the terminal area (TMA).

2.4 Transition from Flight Levels to Altitudes and vice versa

- 2.4.1 The vertical position of aircraft when at or below the transition altitude shall be expressed in terms of altitude; whereas such position at or above the transition level shall be expressed in terms of flight levels. While passing through the transition layer, the vertical position shall be expressed in terms of flight level when ascending, and in terms of altitude when descending.

2.5 Take-Off and Climb

- 2.5.1 Prior to taking off, one altimeter shall be set on the latest QNH altimeter setting for the aerodrome.
- 2.5.2 During climb to and while at the transition altitude, references to the vertical position of the aircraft as contained in air ground communications shall be expressed in terms of altitude.
- 2.5.3 On penetrating the transition altitude, the reference for vertical position of the aircraft shall be changed from altitude (QNH) to flight levels (1013.2Mb), and thereafter, vertical position shall be expressed in terms of flight levels.

2.6 Enroute

- 2.6.1 Vertical Separation
- 2.6.1.1 During enroute flight at or below the transition altitude, an aircraft shall be flown at altitudes and references to the vertical position of the aircraft as contained in air ground communications and shall be expressed in terms of altitudes.
- 2.6.1.2 During enroute flight at or above the transition levels or the lowest usable flight level, whichever is applicable, an aircraft shall be flown at flight levels and references to the vertical position of the aircraft as contained in air ground communications and shall be expressed in terms of flight levels.
- 2.6.2 Terrain Clearance
- 2.6.2.1 Where adequate QNH altimeter reports are available, the latest and most appropriate reports shall be used for assessing terrain clearance.
- 2.6.2.2 Where the adequacy of terrain clearance cannot be assessed with an acceptable degree of accuracy by means of the QNH reports available or forecast lowest mean sea level pressure, other information shall be obtained for checking the adequacy of terrain clearance.

2.7 Approach and Landing

- 2.7.1 Prior to commencing the initial approach to an aerodrome the number of the transition level shall be obtained.

Note: The transition level is normally obtained from the appropriate air traffic services unit.

- 2.7.2 Prior to descending below the transition level, the latest QNH altimeter setting for the aerodrome shall be obtained.

Note: The latest QNH altimeter setting for the aerodrome is normally obtained from the appropriate ATS unit.

- 2.7.3 On descending below the transition level, the reference for the vertical position shall be changed from flight levels (1013.2Mb) to altitudes (QNH), and thereafter, the vertical position of the aircraft shall be expressed in terms of altitudes.

Note: This does not preclude a pilot using a QFE setting for the terrain clearance purposes during the final approach to the runway in accordance with 2.7.5.

- 2.7.4 The reference for the vertical position may be changed from flight levels (1013.2Mb) to altitudes (QNH) above the transition level, when cleared to do so by the appropriate ATS unit after approach clearance has been issued and the descent to land is commenced provided that level flight above the transition altitude is not indicated or anticipated.

- 2.7.5 When an aircraft which has been given a clearance as number one to land is completing its approach using QFE, the vertical position of the aircraft shall be expressed in terms of height above the aerodrome datum used for the obstacle clearance limit during that portion of its flight for which the QFE may be used.

2.8 Missed Approach

- 2.8.1 The altimeter setting used while completing a missed approach procedure will be dependent upon whether or not the procedure can be carried out below the transition altitude, and in any event, they should be consistent with the relevant procedures contained in subsections 2.5, 2.6 and 2.7.

2.9 Established Transition Altitudes and Transition Levels

All transition altitudes and levels within the Manila FIR are set uniformly at 11000FT and FL130, respectively.

Note: Aircraft transiting through TMA's at 11000FT or below shall set their altimeter to the QNH of the areas upon entry and revert to standard setting upon departure.

3. Description of Altimeter Setting Region (not applicable)

4. Procedures Applicable to Operators (including pilots)

4.1 Flight Planning

- 4.1.1 The levels to be conducted shall be specified in a flight plan:

- a. In terms of flight levels (FL) if the flight is to be conducted at or above the transition level or the lowest usable flight level, whichever is applicable;
- b. In terms of altitude if the flight is to be conducted at or below the transition altitude in accordance with subsection 2.2

- 4.1.2 The altitudes or flight levels selected for flight:
- a. Should ensure adequate terrain clearance at all points along the route to be flown;
 - b. Should satisfy air traffic control requirements; and
 - c. Should be compatible with the application of the table of cruising levels in Appendix 1 ENR 1.2, and Appendix 2 ENR 1.3.

5. Table of Cruising Levels

Refer to ENR 1.2 Appendix 1 and ENR 1.3 Appendix 2.